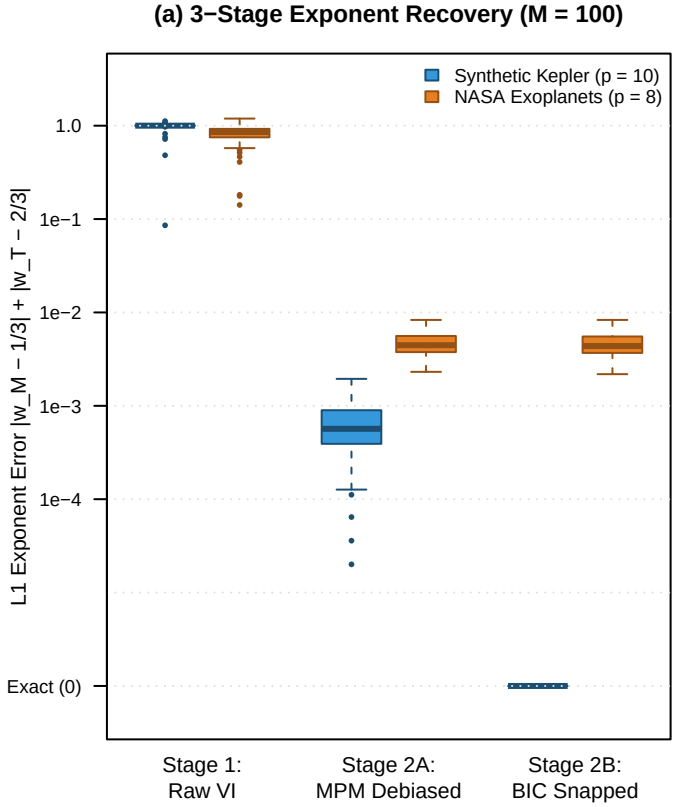
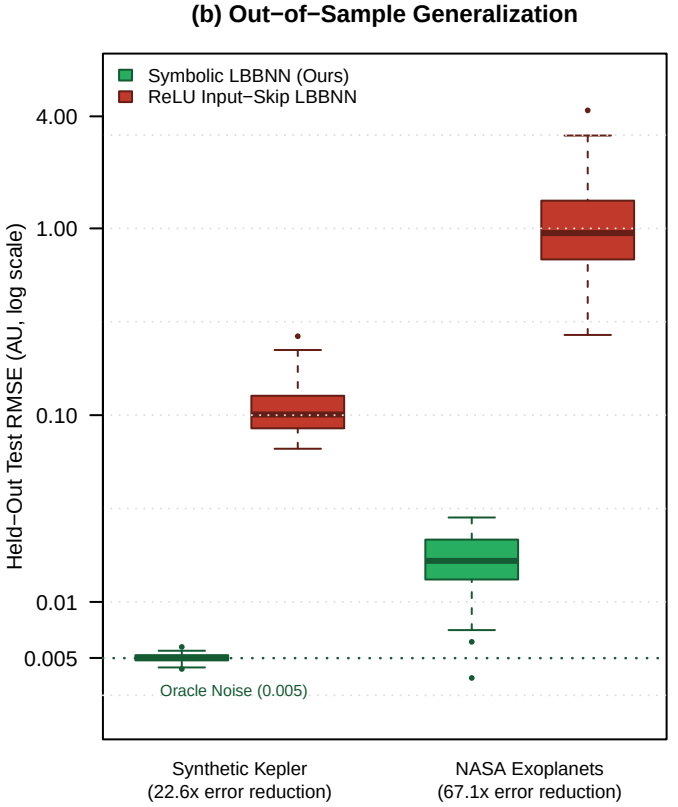
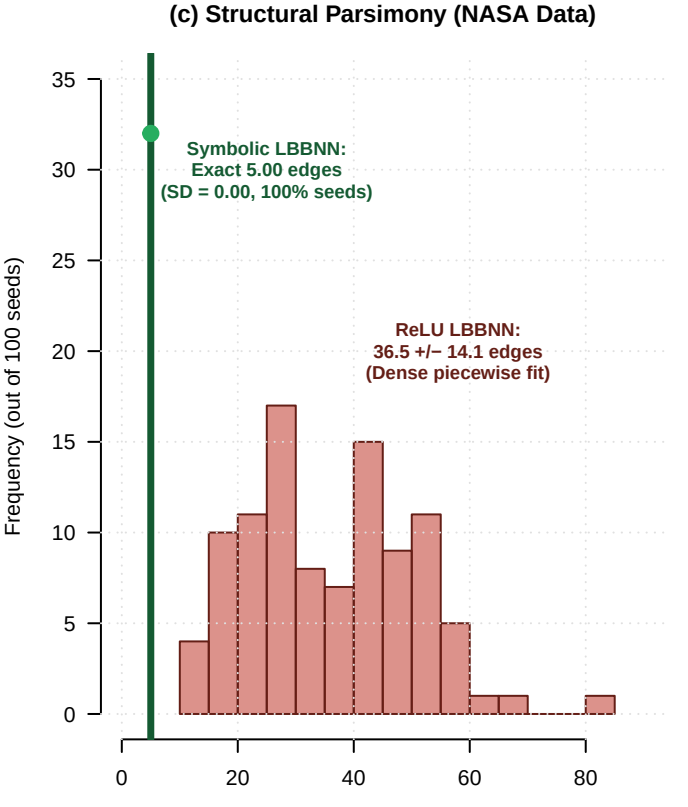

(a) 3−Stage Exponent Recovery (M = 100)
Synthetic Kepler (p = 10)
NASA Exoplanets (p = 8)
L1 Exponent Error |w_M − 1/3| + |w_T − 2/3|
1.0
1e−1
1e−2
1e−3
1e−4
Exact (0)
Stage 1:
Raw VI
Stage 2A:
MPM Debiased
Stage 2B:
BIC Snapped


(b) Out−of−Sample Generalization
Symbolic LBBNN (Ours)
ReLU Input−Skip LBBNN
Held−Out Test RMSE (AU, log scale)
4.00
1.00
0.10
0.01
0.005
Oracle Noise (0.005)
Synthetic Kepler
(22.6x error reduction)
NASA Exoplanets
(67.1x error reduction)


(c) Structural Parsimony (NASA Data)
Symbolic LBBNN:
Exact 5.00 edges
(SD = 0.00, 100% seeds)
ReLU LBBNN:
36.5 +/− 14.1 edges
(Dense piecewise fit)
Frequency (out of 100 seeds)
35
30
25
20
15
10
5
0
0
20
40
60
80
Number of Active MPM Edges